

Proving the Reflection Principle of Brownian Motion using Lévy's Characterization

Name:Jia Dingkang **Student ID:**202300091035 **Department:** Statistics Class

目录

1	Statement of the Reflection Principle for Brownian Motion	2
2	Proof of the Proposition	2
2.1	Proof of Lemma	2
2.2	Proof of the Reflection Principle of Brownian Motion	2
3	Postscript	3

1 Statement of the Reflection Principle for Brownian Motion

Let $\tilde{B}_t = (\tilde{B}_t)_{t \geq 0}$ be defined on the probability space $(\Omega, \mathcal{F}, P)$ (where $B = (B_t)_{t \geq 0}$ is a standard Brownian motion on $(\Omega, \mathcal{F}, P)$) by:

$$\tilde{B}_t = \begin{cases} B_t, & t < T_a \\ 2a - B_t, & t \geq T_a \end{cases}$$

where T_a (the **stopping time**) is defined as:

$$T_a = \begin{cases} \inf\{t \geq 0 : B_t \geq a\}, & a > 0 \\ \inf\{t \geq 0 : B_t \leq a\}, & a < 0 \end{cases}$$

(which simplifies to $\inf\{t \geq 0 : B_t = a\}$).

2 Proof of the Proposition

Next, we present the lemmas to be used:

Lemma (Lévy's Characterization of Brownian Motion)

Let $B = (B_t)_{t \geq 0}$ be a continuous process on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P)$. The following are equivalent:

1. B is a standard Brownian motion.
2. B is a continuous local martingale, and $(B_t^2 - t)_{t \geq 0}$ is a martingale.

Lemma

Suppose M is a continuous local martingale with $M_0 = 0$. Then the following assertions are equivalent:

1. For any $t > 0$, $\mathbb{E}[\langle M, M \rangle_t] < +\infty$;
2. M is a martingale, and for any $t > 0$, $\mathbb{E}[M_t^2] < +\infty$.

2.1 Proof of Lemma

Refer to the proofs of **Proposition 4.2.6** and **Lemma 5.2.1** in *Fundamentals of Stochastic Processes*.

2.2 Proof of the Reflection Principle of Brownian Motion

We first verify the **continuity** of $\tilde{B}$:

Fix a sample path ω ; $B_t(\omega)$ is a constant for this ω . For $t < T_a(\omega)$, $\tilde{B}_t(\omega) = B_t(\omega)$ (continuous in t). For $t > T_a(\omega)$, $\tilde{B}_t(\omega) = 2a - B_t(\omega)$ (also continuous in t). At $t = T_a(\omega)$: The left limit of $\tilde{B}_t(\omega)$ (as $t \rightarrow T_a^-$) is $B_{T_a}(\omega) = a$. The right limit of $\tilde{B}_t(\omega)$ (as $t \rightarrow T_a^+$) is $2a - B_{T_a}(\omega) = a$.

Thus $\tilde{B}_t(\omega)$ is continuous at $t = T_a(\omega)$, so $\tilde{B}_t(\omega)$ is continuous in t for this ω . Since the set of ω where $B_t(\omega)$ is discontinuous is negligible (by continuity of Brownian motion), $\hat{B}$ has continuous sample paths almost surely.

$\tilde{B}_t = B_t \mathbf{1}_{\{t < T_a\}} + (2a - B_t) \mathbf{1}_{\{t \geq T_a\}}$. This can be rewritten as:

$$\tilde{B}_t = B_t + 2(a - B_t) \mathbf{1}_{\{t \geq T_a\}}.$$

Consider the process $B_t - 2 \int_0^t \mathbf{1}_{\{s \geq T_a\}} dB_s$: When $t < T_a$:

$$B_t - 2 \int_0^t \mathbf{1}_{\{s \geq T_a\}} dB_s = B_t.$$

When $t \geq T_a$:

$$B_t - 2 \int_0^t \mathbf{1}_{\{s \geq T_a\}} dB_s = B_t - 2 \int_{T_a}^t dB_s = B_t - 2(B_t - B_{T_a}) = 2B_{T_a} - B_t = 2a - B_t.$$

Thus, we conclude:

$$\tilde{B}_t = B_t - 2 \int_0^t \mathbf{1}_{\{s \geq T_a\}} dB_s.$$

The process $\int_0^t \mathbf{1}_{\{s \geq T_a\}} dB_s$ is a local martingale.

Since

$$\mathbb{E} \left[\int_0^t (\mathbf{1}_{\{s \geq T_a\}})^2 ds \right] < +\infty \quad \text{for } t < +\infty,$$

it follows that $\int_0^t \mathbf{1}_{\{s \geq T_a\}} dB_s$ is a martingale.

Therefore, $(\tilde{B}_t)_{t \geq 0}$ is a martingale.

We can also express $\tilde{B}_t$ as:

$$\tilde{B}_t = \int_0^t (1 - 2\mathbf{1}_{\{s \geq T_a\}}) dB_s.$$

The quadratic variation of $\tilde{B}$ is:

$$\langle \tilde{B}, \tilde{B} \rangle_t = \int_0^t (1 - 2\mathbf{1}_{\{s \geq T_a\}})^2 ds = \int_0^t 1 ds = t.$$

By **Lemma** (Lévy's Characterization), $\hat{B}$ is a **standard Brownian motion** on $(\Omega, \mathcal{F}, P)$.

3 Postscript

I would like to express my sincere gratitude to Professor Wang for his dedicated instruction in the "Stochastic Processes" course this semester. Stochastic processes theory is abstract and

challenging. I used to feel intimidated by the numerous and complex symbols in the textbook. However, through Professor Wang's thorough explanations in class and my repeated reflections after class, I gradually gained insight into the clear internal logic of this course. Your concrete examples not only gave us a more intuitive understanding of the classroom knowledge but also allowed us to appreciate the unique charm of this subject. In short, through studying this course, I have not only firmly mastered professional knowledge but also developed a rigorous academic attitude and logical thinking abilities. This learning experience will serve as a solid foundation for my future research and studies. Once again, I extend my heartfelt thanks to Professor Wang! Wish you every success in your work!

References

- [1] Revuz, D., & Yor, M. (1999). *Continuous Martingales and Brownian Motion* (3rd ed.). Berlin: Springer. (Chapter VI)
- [2] 王汉超. 随机过程基础